HW1

Saturday, 14 January 2023 9:24 PM

Problem 2.1

Linearly independent reactions → No one stoichiometric equation can be derived from the others by a linear combination

(c) $A_i \rightarrow A_j$; $i, j = 1, 2, \ldots, n$, $i \neq j$

Linearly Ind. set:

$A_1 \rightarrow A_2$

$A_2 \rightarrow A_3$

$A_3 \rightarrow A_4$

$A_4 \rightarrow A_5$

⋮

$(n-1)$

$$\begin{bmatrix} -1 & 1 & 0 & 0 & \cdots & 0 \\ 0 & -1 & 1 & 0 & \cdots & 0 \\ 0 & 0 & -1 & 1 & \cdots & 0 \\ 0 & 0 & 0 & -1 & 1 & \cdots \\ \vdots & & & & \ddots & \\ & & & & -1 & 1 \end{bmatrix}$$ (n columns, n−1 rows)

↳ $A_1 \rightarrow A_1$ is not possible (since $i \neq j$)

↳ $A_1 \rightarrow A_2$, $A_2 \rightarrow A_1$ } These form linearly dependent set since differ by a factor of −1.

∴ (n−1) Linearly independent reactions as shown in the set.

Uniqueness: The given set is not unique since other linearly independent sets can also be constructed

1) $A_1 \rightarrow A_2$, $A_2 \rightarrow A_3$, ... $A_{n-1} \rightarrow A_n$

2) $A_n \rightarrow A_1$, $A_n \rightarrow A_2$, ... $A_n \rightarrow A_{n-1}$

(d) $A_1 = NaCl$

$A_2 = H_2SO_4$

$A_3 = Na_2SO_4$

$A_4 = HCl$

$A_5 = C$

$A_6 = Na_2S$

$A_7 = CO$

$A_8 = CaCO_3$

$A_9 = Na_2CO_3$

$A_{10} = CaS$

$$\begin{cases} 2A_1 + A_2 \rightarrow A_3 + 2A_4 \\ A_3 + 4A_5 \rightarrow A_6 + 4A_7 \\ A_6 + A_8 \rightarrow A_9 + A_{10} \end{cases}$$

$$\begin{bmatrix} -2 & 1 & 1 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & -4 & 1 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 1 & 1 \end{bmatrix}$$

rank = 3

10 reactants & 3 reactions

↳ Full rank matrix since no two equations can form the other

↳ 3 independent reactions and all three form a unique set

m × n

8 × 9

(e) $A_1 = C_3H_8$

$A_2 = C_2H_4$

$A_3 = CH_4$

$A_4 = C_3H_6$

$A_5 = H_2$

$A_6 = C_2H_6$

$A_7 = C_2H_2$

$A_8 = C_4H_8$

$A_9 = C_4H_6$

$A_1 \rightarrow A_2 + A_3$

$A_1 \rightarrow A_4 + A_5$

$A_1 + A_2 \rightarrow A_6 + A_4$

$2A_4 \rightarrow 3A_2$

$A_4 \rightarrow A_7 + A_3$

$A_4 + A_6 \rightarrow A_8 + A_3$

$A_6 \rightarrow A_2 + A_5$

$A_2 + A_7 \rightarrow A_9$

9 reactants, 8 reactions

$$\begin{bmatrix} -1 & +1 & +1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & +1 & +1 & 0 & 0 & 0 \\ -1 & -1 & 0 & +1 & 0 & +1 & 0 & 0 \\ 0 & +3 & 0 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & +1 & +1 & 0 & 0 & +1 & 0 \\ 0 & 0 & +1 & -1 & 0 & -1 & 0 & +1 \\ 0 & +1 & 0 & 0 & +1 & -1 & 0 & 0 \\ 0 & +1 & 0 & 0 & 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \\ A_5 \\ A_6 \\ A_7 \\ A_8 \\ A_9 \end{bmatrix}$$

rank = 7

(i.e. 7 pivots)

∴ 7 linearly independent reactions

Set:

All reactions excluding reaction 7

∴ $A_1 \rightarrow A_2 + A_3$

$A_1 \rightarrow A_4 + A_5$

$A_1 + A_2 \rightarrow A_6 + A_4$

$2A_4 \rightarrow 3A_2$

$A_4 \rightarrow A_7 + A_3$

$A_4 + A_6 \rightarrow A_8 + A_3$

$A_2 + A_7 \rightarrow A_9$

Uniqueness: The given set is not unique since one of reaction is linearly dependent and therefore can be expressed in a different set.

Problem 2.3

Matrix A → stoichiometric matrix

Differential equation of Multiple reactions in batch

$$\frac{d\xi_i}{dt} = r_i - \xi_i \frac{d\ln V}{dt} \quad \text{... Eq 1}$$

If $\text{rank}(A) = r < m$ (m: no. of reactions)

∴ There are r linearly independent reactions given by the above equation (Eq 1) where $i = 1, 2, 3 \ldots, r$

Additionally, since volume is changing there is additional volume equation given by:

$$\frac{d\ln V}{dt} = \sum_{i=1}^{m} r_i \sum_{j=1}^{s} \alpha_j \bar{V}_j$$

∴ Minimum no. of differential eq: $r+1$